

Comparing Distributions and Compound Events

Answer Key

Grades 6–7 Mathematics

Quick Answers

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|--|---|
| 1. 7 | 8. (a) the median = 6, (b) the mean = 8.4, — |
| 2. 8 | 9. the probability from the counts = $\frac{9}{64}$, the probability in lowest terms = $\frac{9}{64}$ |
| 3. 60 | 10. (a) Wednesday's total = 15, (b) the fraction of the week = $1 \cdot \frac{1}{4}$ |
| 4. $1 \cdot \frac{1}{12}$ | 11. $\frac{3}{16}$ |
| 5. (a) range of the first team = 5, (b) range of the second team = 7, — | 12. (a) mean of the first class = 8, (b) mean of the second class = 8, (c) range of the first class = 6, (d) range of the second class = 2, — |
| 6. $1 \cdot \frac{1}{6}$ | |
| 7. (a) the busiest day's total = 19, (b) the quietest day's total = 6, (c) the difference = 13 | |
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What is verified

9 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 5, 8, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 6–7 Mathematics

6.SP.B.5 – *problems 1, 2, 3, 5, 7, 8, 10, 12*

7.SP.C.8 – *problems 4, 6, 9, 11*

Difficulty 1–5 of 5 across 24 tagged checks.

1. Falcons scored 4, 6, 7, 9, 9. Find the mean.

Solution: Add the five scores to get 35, then share that total equally between the five games: $35 \div 5$. 7

2. Herons scored 5, 5, 8, 10, 12. Find the mean.

Solution: The five scores total 40, and $40 \div 5$ gives the fair share per game. 8

3. Using Display A, how many sandwiches were sold in the whole week?

Solution: Read each bar against the scale and add all five days: $12 + 8 + 15 + 6 + 19$. 60

4. A coin is flipped and a number cube is rolled. Probability of heads and a five?

Solution: The two stages happen together, so the total number of outcomes is the coin's two results times the cube's six, giving twelve equally likely pairs. Exactly one of those pairs is heads with a five. $\frac{1}{12}$

5. Ranges of the two teams, and which was more spread out.

Solution (a): Falcons: largest minus smallest, $9 - 4$. $\frac{5}{7}$

Solution (b): Herons: largest minus smallest, $12 - 5$. $\frac{7}{7}$

Solution (c) — instructor-judged. The Herons are the more spread out: their range is the larger of the two, so their game-to-game scoring varied more, while the Falcons' narrower range means a more predictable performance. A full-credit answer names the team and quotes both range values as the evidence, rather than saying only that one is bigger.

6. Two number cubes are rolled. Probability the total is seven?

Solution: Six faces on the first cube times six on the second gives thirty-six equally likely pairs. The pairs totalling seven are one-and-six, two-and-five, three-and-four and their three reverses — six of them. Six out of thirty-six reduces. $\frac{1}{6}$

7. Using Display A: busiest day, quietest day, and the difference.

Solution (a): The tallest bar is Friday. $\frac{19}{6}$

Solution (b): The shortest bar is Thursday. $\frac{6}{13}$

Solution (c): Subtract the smaller from the larger. $\frac{13}{13}$

8. Umbrellas sold: 3, 5, 6, 8, 20. (a) median, (b) mean, (c) which is more typical?

Solution (a): The values are already in order, and the middle one of five is the third. $\frac{6}{8.4}$

Solution (b): The five values total 42, and $42 \div 5$ gives the mean. $\frac{8.4}{8.4}$

Solution (c) — instructor-judged. The median is the better report. One day sold far more umbrellas than the rest, and that single value drags the mean up above four of the five days — so the mean describes no typical day at all, while the median sits among the ordinary ones. Full credit requires naming that unusually large value as the reason.

9. Bag of 3 red and 5 blue marbles, drawn twice *with replacement*. Probability both are red?

Solution: There are eight marbles, so a single draw is red with probability three eighths. The marble goes back, so the second draw has exactly the same chance. The two stages multiply: three eighths of three eighths. $\frac{9}{64}$

10. Using Display A: Wednesday's total, and its fraction of the week.

Solution (a): Read the Wednesday bar against the scale. $\frac{15}{15}$

Solution (b): The week's total from problem 3 is sixty, so the fraction is fifteen out of sixty, which reduces by dividing both parts by fifteen. $\frac{1}{4}$

11. Spinner numbered one to four, spun twice. Probability the total exceeds six?

Solution: Four results on the first spin times four on the second gives sixteen equally likely pairs. Totals above six come from three-and-four, four-and-three, and four-and-four — three cells of the table. The fraction three sixteenths does not reduce. $\frac{3}{16}$

12. Challenge. Class One 6, 7, 7, 8, 12; Class Two 7, 8, 8, 8, 9. Means, ranges, and which class was steadier.

Solution (a): Class One totals 40 over five students.

8

Solution (b): Class Two also totals 40 over five students.

8

Solution (c): Class One: $12 - 6$.

6

Solution (d): Class Two: $9 - 7$.

2

Solution (e) — instructor-judged. Class Two was the steadier. The two means are identical, so centre alone cannot separate the classes at all — a point worth drawing out, because it is the whole reason spread is reported. The ranges decide it: Class Two's scores span a third as much as Class One's, whose top score sits well clear of the rest. Full credit requires both moves: noticing that the centres tie, and quoting the two ranges as the deciding evidence.